

E-Rate Internet Proposal Analysis

January, 2019

Executive Summary

The Brecksville-Broadview Heights School District has reviewed proposals for Internet access from seven providers. In addition to the current bandwidth need, the proposals were considered in light of a projected increase in bandwidth demand over the next three years. Also considered were the reputations and previous experience with the providers as well as the internal cost of switching providers.

Based on this analysis, it is recommended that the district enter into a three-year agreement with Connect Ohio for Internet access at a contract amount of \$37,000 per year for 1 gb. The actual cost to the district after discounts and subsidies will be \$11,400. This represents a cost decrease of 46% compared to the current contract. Additionally, this contract ensures that a cost-effective structure is in place to meet the projected bandwidth needs as they increase over the next three years.

Background and Needs Analysis

As part of the federal e-Rate modernization order in 2014, the Federal Communications Commission adopted bandwidth guidelines for schools that established a target of 1 mb per student by the 2017-2018 school year.¹ Additionally, the Southwest Ohio Computer Association (SWOCA), an organization that provides Internet access and technology services to schools in the Cincinnati area, created a bandwidth estimation tool in 2018 based on actual bandwidth use by its member schools.² This model includes such factors as enrollment, number of devices in use, types of devices and deployment models, and filtering policies to develop guidelines for district bandwidth needs.

According to the FCC model, the Brecksville-Broadview Heights Schools, with 3,800 students, should have a goal of 3,800 mb, or 3.8 gb. Using the SWOCA model and characteristics of the district, the estimated need for 2017-18 was 558 mb, with 50% growth per year (up to 2.8 gb by 2021).

Additionally, the Management Council of the Ohio Educational Computing Network (MCOECN) reported in 2012 that school bandwidth needs tend to increase, on average, by about 50% per year.³ This is consistent with SWOCA's estimates (57% per year over the last five years), as well as the

¹ "K-12 Bandwidth Goals." Education SuperHighway. January 2019. <<http://bit.ly/2sOVfo0>>

² "District Bandwidth Forecast Model" Southwest Ohio Computer Association. February 2018. <<http://bit.ly/2F0tD4Q>>

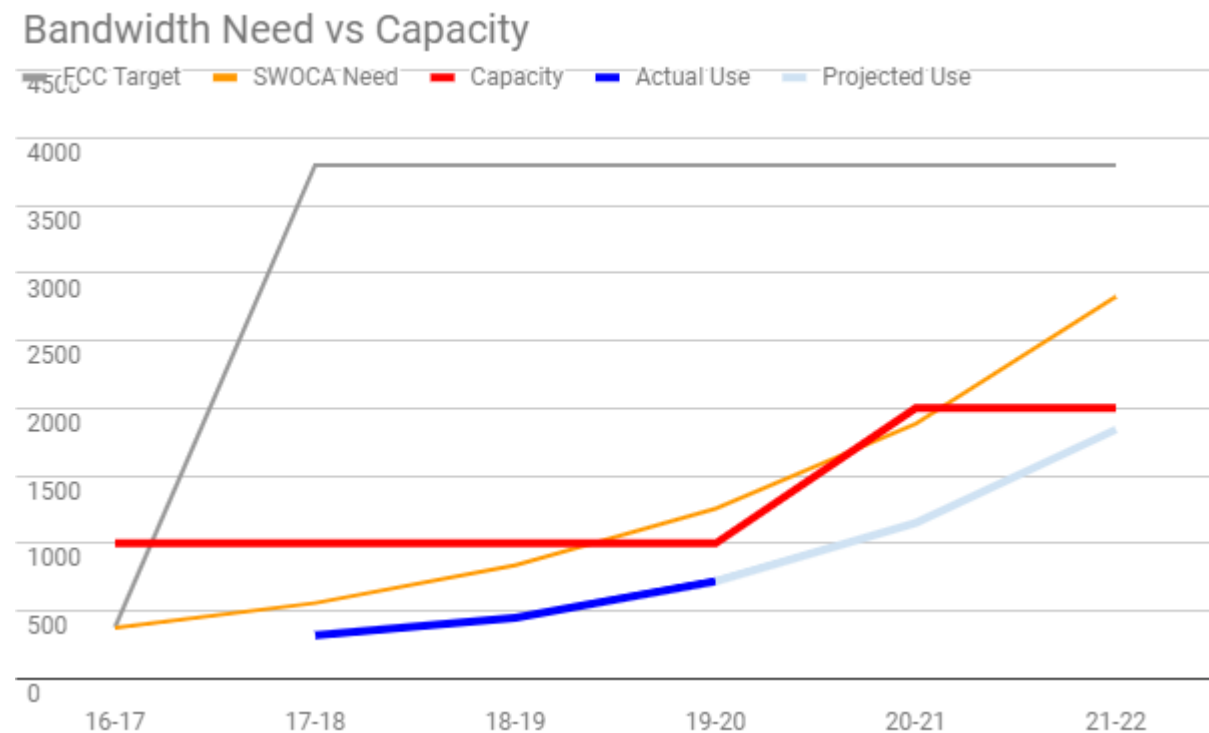
³ "Ohio K-12 Network Upgrade Analysis." MCOECN. Management Council of the Ohio Educational Computing Network, 8 Mar. 2012. Web. 27 Mar. 2017. <<http://www.mcoecn.org/wp-content/uploads/2012/06/ODLTF-30812-FINAL.pdf>>.

school district's experience over the past 20 years. That is, if the district is using 1.0 gb in a given year, it is likely to need 1.5 gb the following year, 2.3 gb in two years, etc.

Under the current Internet contract, the school district has a 1 gb connection to the Internet. The last upgrade came in the summer of 2016 after several years of significant congestion. This connection is meeting the current needs of the district. While bandwidth use is difficult to measure, reports from the district's service provider suggest that it is using about half of the current 1 gb connection. This is about a year behind the SWOCA estimate and significantly lower than the FCC guideline.

Projecting forward, the district is likely to be fine with the current connection for one year, after which upgrades will be needed. To make an informed decision, the district requested pricing from potential service providers at the 1 gb, 2 gb, 5 gb, and 10 gb levels, with the option to upgrade within the term of the three-year contract. The most likely scenarios will be a need for 1 gb in the first year, 2 gb in the second year, and either 2 gb or 5 gb in the third year.

The following chart shows the FCC and SWOCA bandwidth targets, along with the district's capacity, actual, and projected use over the six year term covered by the current and new contracts:



Evaluation Criteria

While it is important to find cost-effective services, price cannot be the only determining factor in selecting a successful bid. In addition to providing an inexpensive solution, other factors must be considered, such as the vendor's reputation and previous experience with the district or other schools in Ohio. Additionally, there is a cost associated with switching away from the current provider. Because IP addresses would change, it would be necessary to re-configure some equipment and services to ensure uninterrupted access. While these challenges are not insurmountable, they do pose an additional burden on the district.

Complete proposals for this project are evaluated based on the following criteria:

Cost	The total cost of the proposal for the most likely bandwidth scenarios identified by the district (1g/2g/2g and 1g/2g/5g). Non-recurring charges are amortized across the term of the contract and included.	80%
Vendor Review	The assessed ability and willingness of the vendor to provide the needed services in a professional, reliable manner. This includes the district's prior experience working with the company.	15%
Change Burden	The assessed need for the district to make changes to current infrastructure to accommodate changing IP addresses and network topology. Higher scores represent a low change burden.	5%

Proposals

Seven companies submitted a total of 24 proposals to provide Internet access for the district. The large number of proposals was in response to the district's request for pricing at the 1 gb, 2 gb, 5 gb, and 10 gb bandwidth levels.

Of the seven companies, two (Connect Ohio and NeoNet) are Ohio Information Technology Centers, which provide a variety of technology services to schools. Two companies (Spectrum and AT&T) are large, well-known providers with proven track records. The other three (EM3 Networks, Cytranet, and Granite Telecommunications) do not have a significant presence in Ohio.

None of these last three companies are currently providing Internet service to public schools in Ohio. AT&T provides Internet service to two districts, but only East Cleveland uses them as their sole Internet provider. Spectrum is providing Internet service to 12 districts in Ohio, though five of these also use a second provider for part of their bandwidth needs. NeoNet provides Internet service to 31 districts, and they're the exclusive provider for all of them. Connect Ohio serves 22 districts, three of which have secondary providers.

Given the lack of a proven track record for EM3, Cytranet, and Granite in Ohio, it is difficult to recommend their use without significant due diligence, especially with a service as mission-critical as Internet access. If the district were pursuing a redundant Internet connection, it may make more sense to look more closely at these providers. It will be interesting to see if these companies gain additional market share in the coming years and provide attractive alternatives to the existing providers.

Regarding the change burden, any provider that is different from the current one would result in the need for changes. Any services hosted within the district that are accessed from outside the network would need to be adjusted. Additionally, online services, such as electronic databases and media resources, rely on the district's IP address range to authenticate students using the service. Coordination with these providers would also be necessary to prevent an interruption of service. These challenges are not insurmountable, but they are also not trivial. Switching to NeoNet would be somewhat less complicated than the other providers, because they work exclusively with schools and would be able to help streamline these processes.

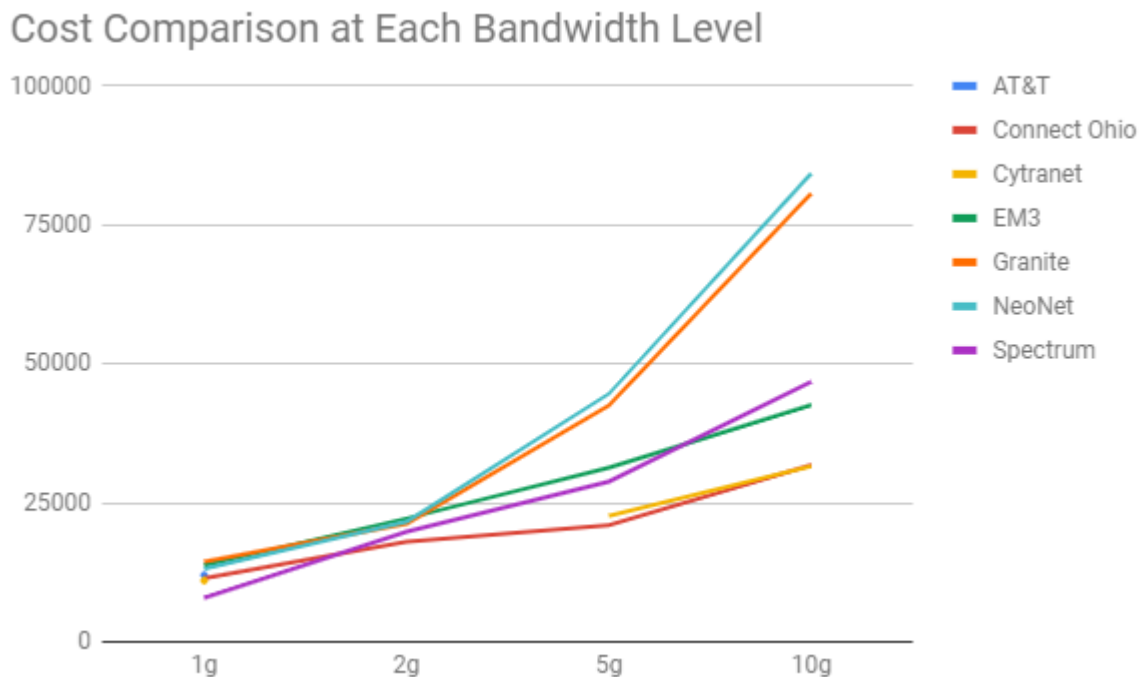
From a cost perspective, the proposals were compared using the district's actual cost. This cost was derived by taking the proposal's monthly cost, amortizing any non-recurring costs across the length of the contract, and deducting the e-Rate and Ohio K-12 Network Subsidy discounts. The Ohio K-12 Network Subsidy is an annual subsidy in the amount of \$10,800 that the district is eligible for if the primary Internet provider is one of the Ohio Information Technology Centers (Connect or NeoNet).

The following table shows the costs for each proposal received. For comparison purposes, the first line shows the current contract.⁴

Comparison of Internet Access Proposals					
Provider	Speed (mb)	Annual Contract Amount	E-rate	K-12 Subsidy	Actual Cost
North Coast Council	1000	\$53,500	-\$21,400	-\$10,800	\$21,300
AT&T	1000	\$19,942	-\$7,977		\$11,965
Connect Ohio	1000	\$37,000	-\$14,800	-\$10,800	\$11,400
Cytranet	1000	\$18,363	-\$7,345		\$11,018
EM3	1000	\$22,942	-\$9,177		\$13,765
Granite	1000	\$24,000	-\$9,600		\$14,400
NeoNet	1000	\$39,840	-\$15,936	-\$10,800	\$13,104
Spectrum	1000	\$13,200	-\$5,280		\$7,920
Connect Ohio	2000	\$48,000	-\$19,200	-\$10,800	\$18,000
EM3	2000	\$36,979	-\$14,792		\$22,188
Granite	2000	\$35,400	-\$14,160		\$21,240
NeoNet	2000	\$54,000	-\$21,600	-\$10,800	\$21,600
Spectrum	2000	\$33,000	-\$13,200		\$19,800
Connect Ohio	5000	\$53,000	-\$21,200	-\$10,800	\$21,000
Cytranet	5000	\$37,749	-\$15,099		\$22,649
EM3	5000	\$52,175	-\$20,870		\$31,305
Granite	5000	\$70,800	-\$28,320		\$42,480
NeoNet	5000	\$92,400	-\$36,960	-\$10,800	\$44,640
Spectrum	5000	\$48,000	-\$19,200		\$28,800
Connect Ohio	10000	\$71,000	-\$28,400	-\$10,800	\$31,800
Cytranet	10000	\$52,653	-\$21,061		\$31,592
EM3	10000	\$70,999	-\$28,400		\$42,600
Granite	10000	\$134,400	-\$53,760		\$80,640
NeoNet	10000	\$158,400	-\$63,360	-\$10,800	\$84,240
Spectrum	10000	\$78,000	-\$31,200		\$46,800

⁴ North Coast Council and Connect Ohio are the same organization. Because the name changed after the current contract was completed in 2016, it is reflected as "North Coast Council" here.

It is interesting to note that the costs are very similar at the 1 gb and 2 gb levels, but they spread out at the higher capacities:



Based on the needs analysis, the district is most likely to need a 1 gb connection in the first year of the contract, followed by a 2 gb connection in the second year, and either a 2 gb or a 5gb connection in the third year. This table shows the three year total cost for both of these scenarios for each provider. AT&T is not included in the list because they did not provide pricing at a level higher than 1 gb. Because Cytranet did not provide pricing at the 2 gb level, their 5 gb pricing is included in both columns here. The Ohio median is included for comparison purposes.

Comparison of Internet Pricing at Different Bandwidth Levels Over 3 Years		
	1g / 2g / 2g	1g / 2g / 5g
Connect	\$47,018	\$50,018
Cytranet	\$56,698	\$56,698
EM3	\$52,704	\$75,384
Granite	\$50,400	\$73,800
Neonet	\$57,600	\$64,800
Spectrum	\$56,340	\$65,457
Ohio Median	\$85,176	\$113,494

Recommendation

The rubric scores for each vendor are listed in the table below. The cost scores were determined by using a linear transformation of the costs for each of the two likely scenarios over the range of those costs, averaging them for each provider, and rounding to the nearest whole number. Because AT&T did not provide the needed pricing for all of the bandwidth levels, their score is equal to the lowest cost score among the other providers.

The vendor scores are based on the school district's experience with the company and its services, along with reviews and opinions of nearby school districts.

The change burden scores are 5 for the current provider, 3 for other ITCs, and 0 for non-school focused Internet providers.

Evaluation Rubric Scores				
Vendor	Cost	Vendor Review	Change Burden	Total
Connect Ohio	83	14	5	102
NeoNet	67	14	3	84
Spectrum	67	12	0	79
Cytranet	73	5	0	78
AT&T	63	10	0	73
Granite	67	5	0	72
EM3	63	5	0	68

Based on this analysis, it is recommended that the district enter into a three-year agreement with Connect Ohio for Internet access at a contract amount of \$37,000 per year for 1gb. The actual cost to the district after discounts and subsidies will be \$11,400. This represents a cost decrease of 46% compared to the current contract. Additionally, this contract ensures that a cost-effective structure is in place to meet the projected bandwidth needs as they increase over the next three years.